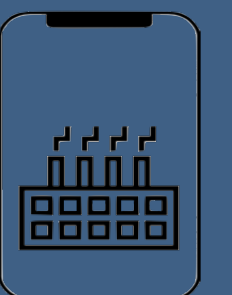
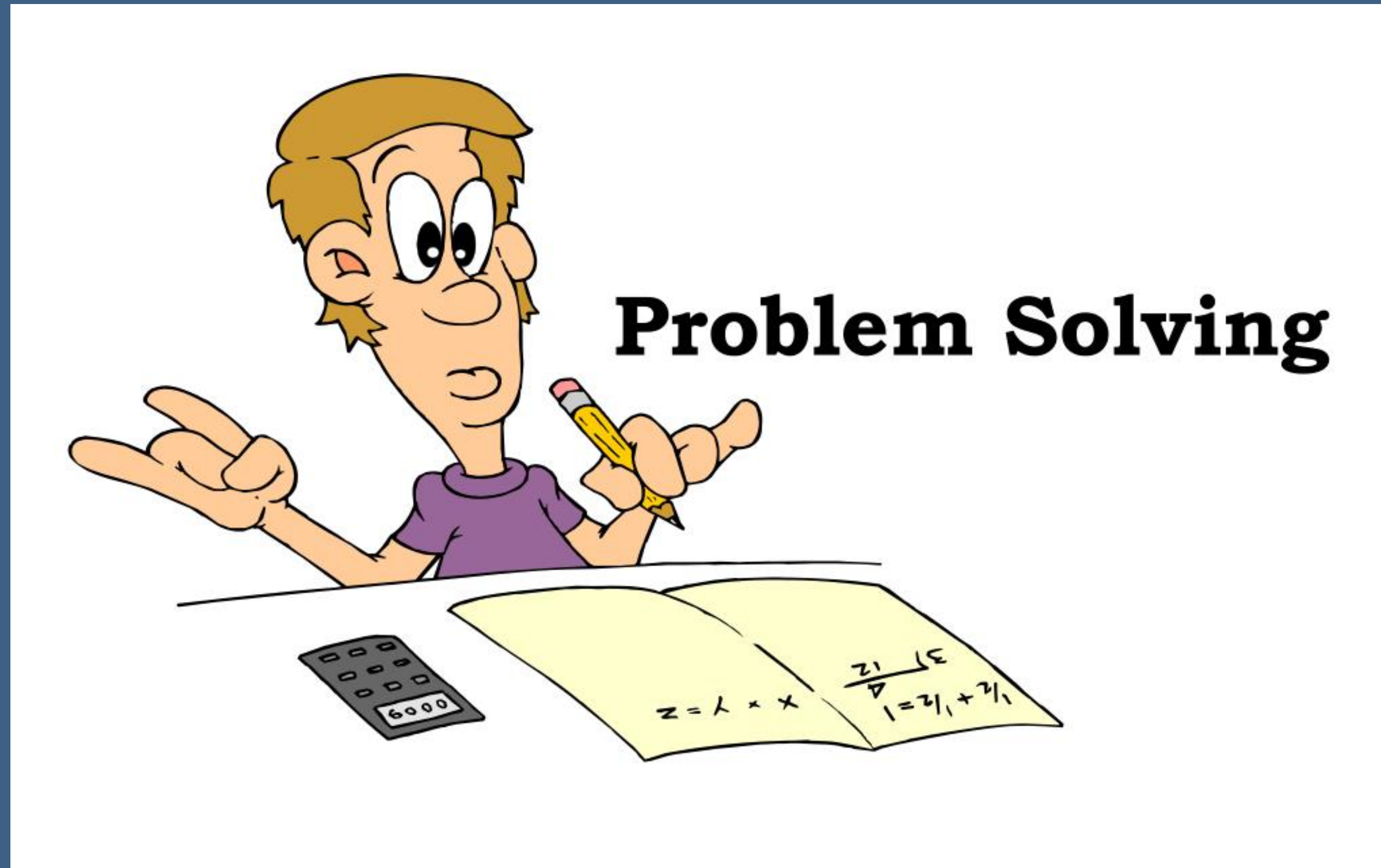


A Recipe for Problem Solving



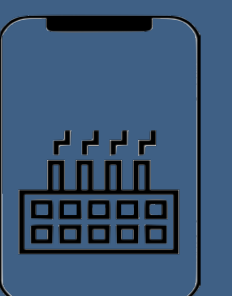
A Recipe for Problem Solving



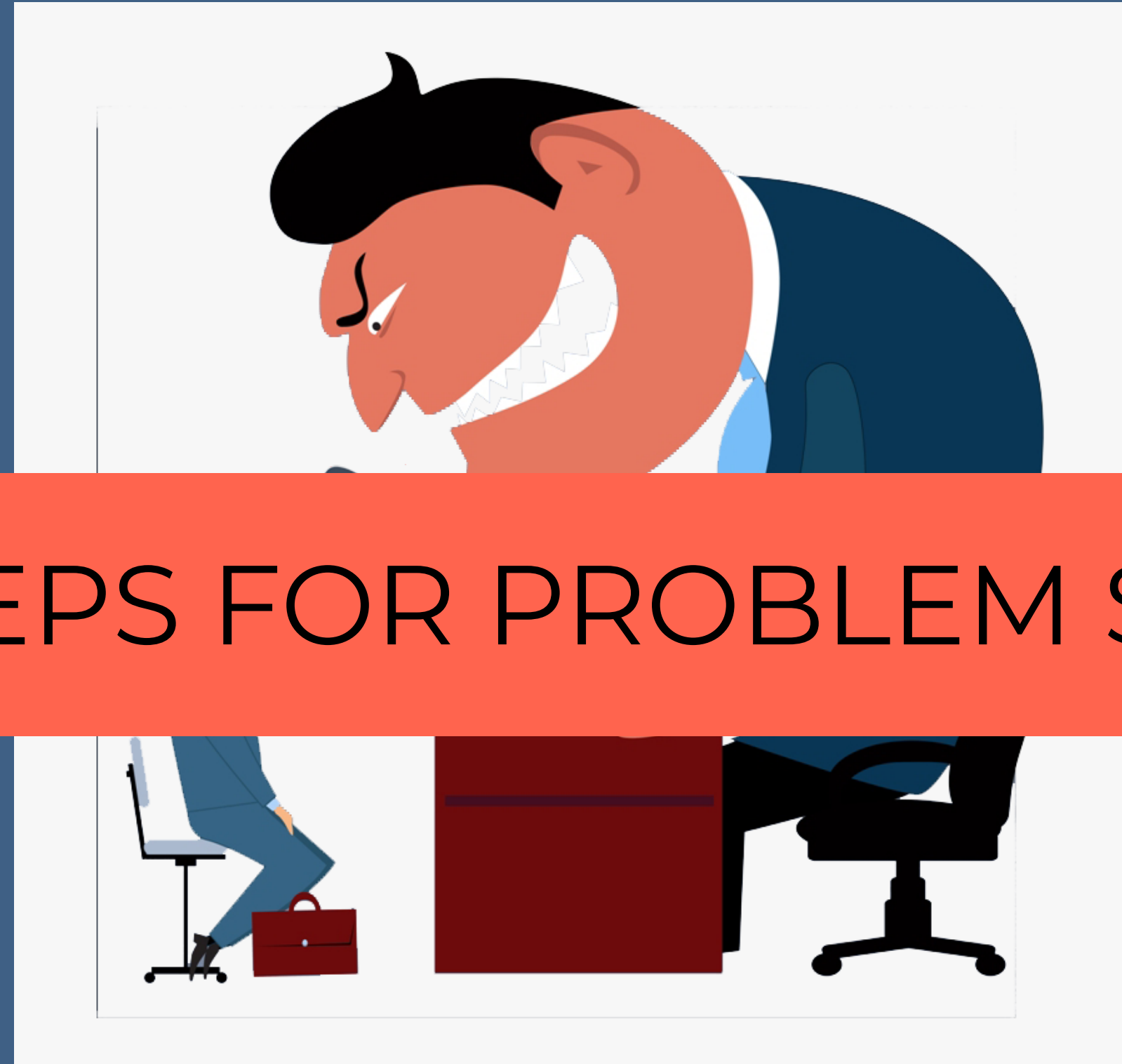
A Recipe for Problem Solving

Algorithm : is a set of steps fo accomplish a certain task

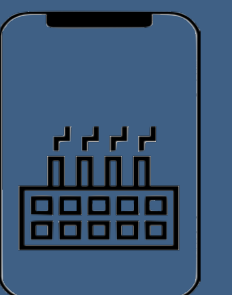
$$(x + a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$



A Recipe for Problem Solving



5 STEPS FOR PROBLEM SOLVING



A Recipe for Problem Solving

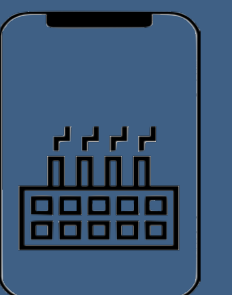
UNDERSTAND THE PROBLEM

EXPLORE EXAMPLES

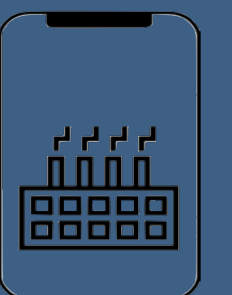
BREAK IT DOWN

SOLVE / SIMPLIFY

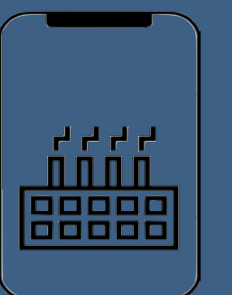
LOOK BACK REFACTOR



Step 1 - UNDERSTAND THE PROBLEM



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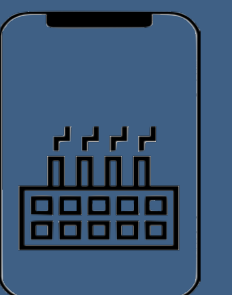
1. Can we restate the problem in our own words?

2. What are the inputs that go into the problem?

3. What are the outputs that come from the problem?

4. Can the outputs be determined from the inputs? In other words do we have enough information to solve this problem?

5. What should I label the important piece of data that are the part of a problem?



Step 1 - UNDERSTAND THE PROBLEM

Write a function that takes two numbers and returns their sum

1. Can we restate the problem in our own words?

Implement addition

2. What are the inputs that go into the problem?

Integer? Float? Or?

3. What are the outputs that come from the problem?

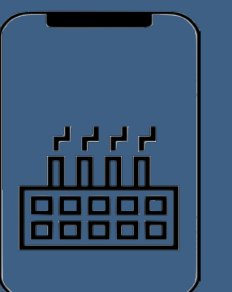
Integer? Float? Or?

4. Can the outputs be determined from the inputs? In other words do we have enough information to solve this problem?

Yes

5. What should I label the important piece of data that are the part of a problem?

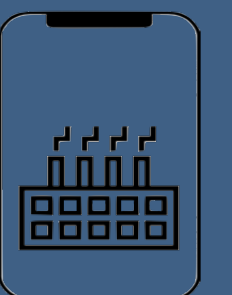
Add, Sum



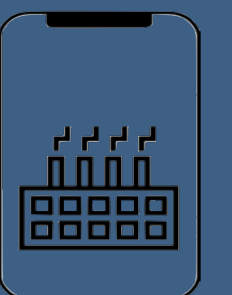
A Recipe for Problem Solving

UNDERSTAND THE PROBLEM

EXPLORE EXAMPLES



Step 2 - EXPLORE EXAMPLES



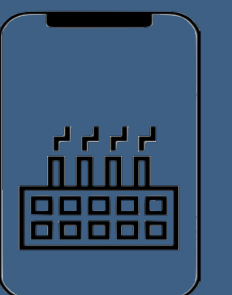
Step 2 - EXPLORE EXAMPLES

1. Start with simple examples

2. Progress to more complex examples

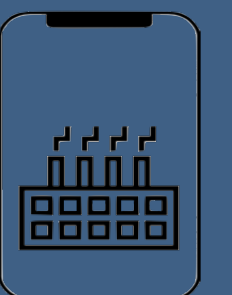
3. Explore examples with empty

4. Explore the examples with invalid inputs



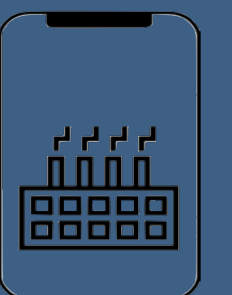
Step 2 - EXPLORE EXAMPLES

Write a function with takes in a string and returns count of each character in the string



Step 3 - BREAK IT DOWN

Write out the steps that you need to take

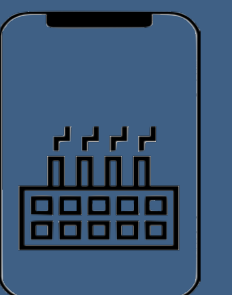


SOLVE / SIMPLIFY

Solve the Problem

If you cannot...

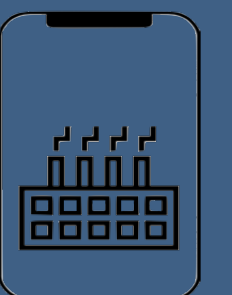
Simplify the Problem



SOLVE / SIMPLIFY

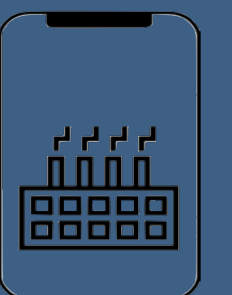
Simplify the Problem

- Find the core difficulty
- Temporarily ignore that difficulty
- Write a simplified solution
- Then incorporate that difficulty



LOOK BACK REFACTOR

- Can we check the result?
- Can we drive the result differently?
- Can we understand it at a glance?
- Can we use the result or method for some other problem?
- Can you improve the performance of your solution?
- How other people solve this problem?



Summarize

UNDERSTAND THE PROBLEM

EXPLORE EXAMPLES

BREAK IT DOWN

SOLVE / SIMPLIFY

LOOK BACK REFACTOR

